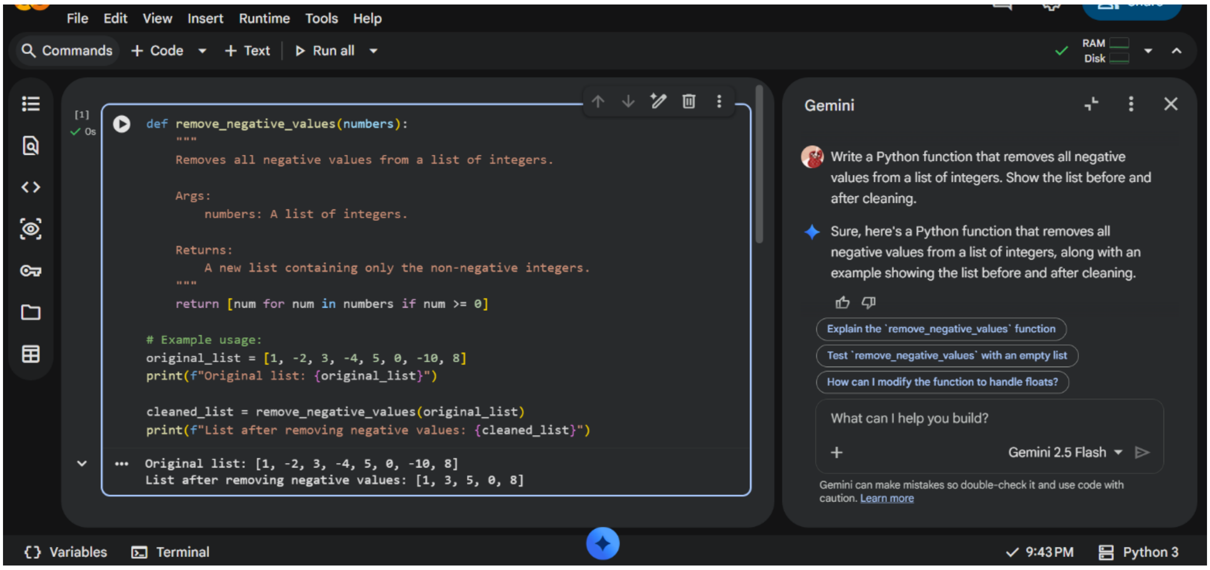
N. Shree Deepthi Batch – 37

AI-ASSISTED CODING ASSIGNMENT 2



Task 1

Prompt Used:

Write a Python function that removes all negative values from a list of integers.

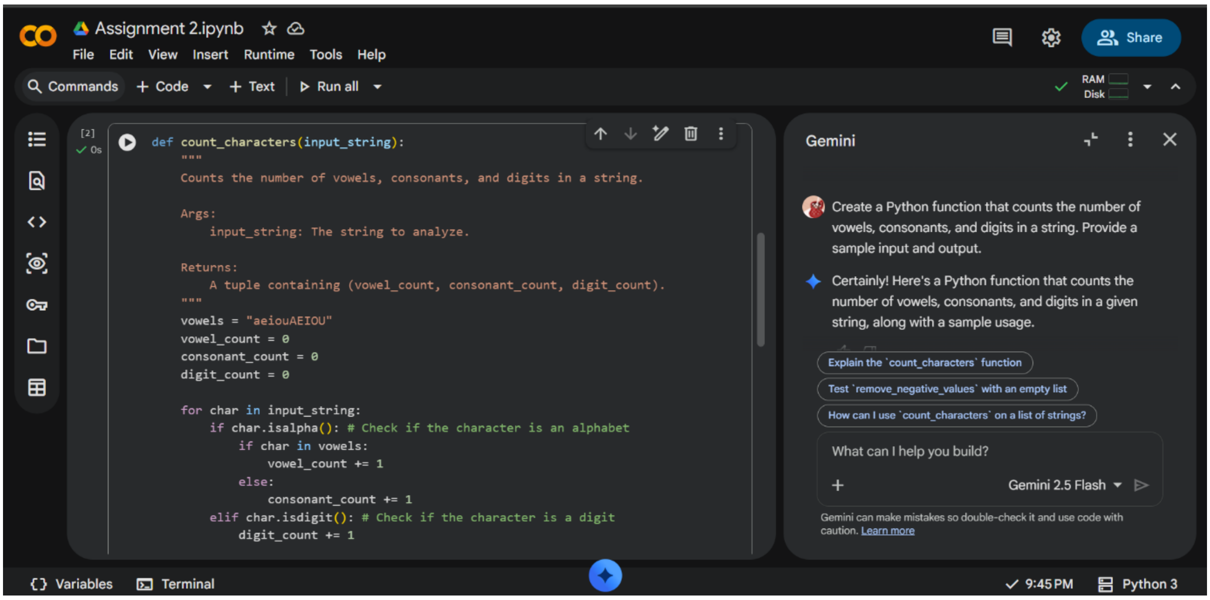
Output:

Before: [10, -3, 5, -8, 12, 0]

After: [10, 5, 12, 0]

Understanding:

Gemini generated a clean and efficient function using lists. The logic was easy to understand and it required no need for manual correction.



Task 2

Prompt Used :

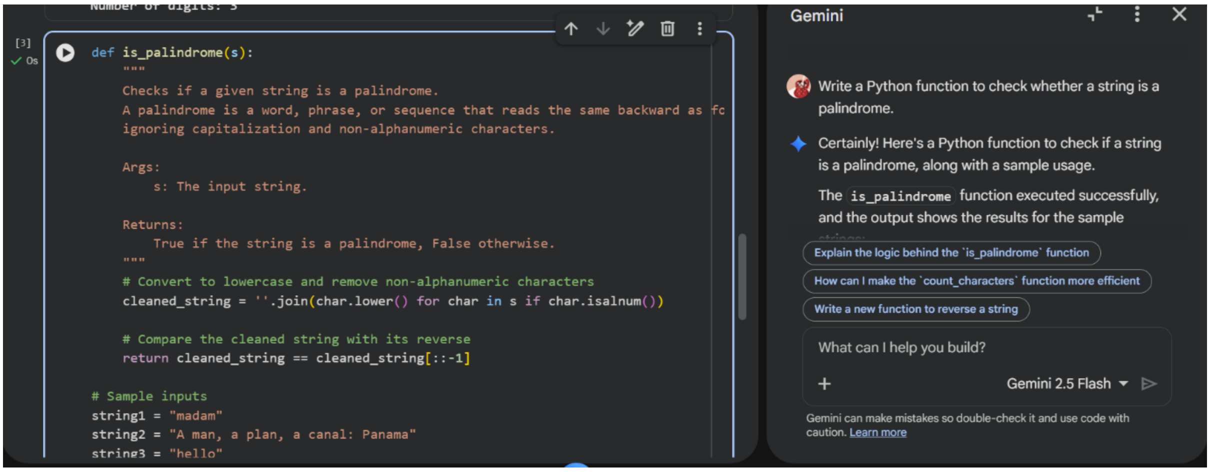
Create a Python function that counts the number of vowels, consonants, and digits in a string.

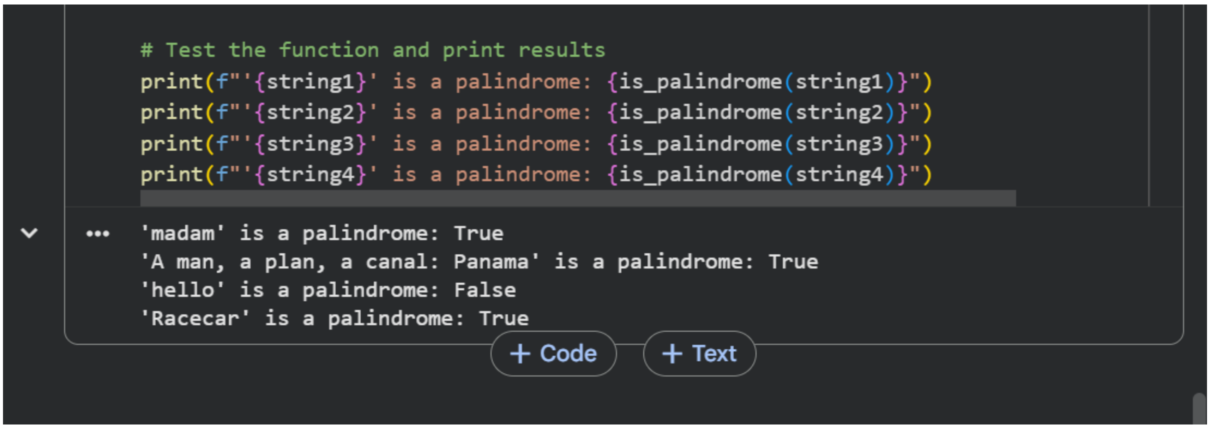
Sample Input:
Hello123

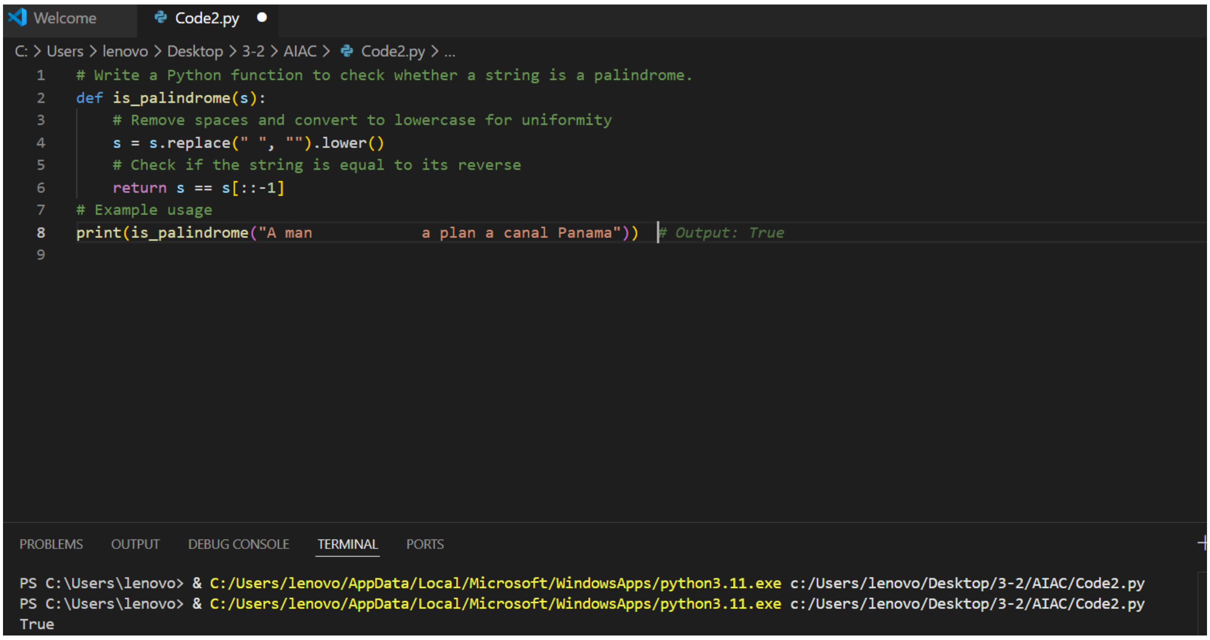
Output:
Vowels: 2
Consonants: 3
Digits: 3

Student Comment:

Gemini clearly handled character classification using built-in Python methods like isalpha() and isdigit() methods.







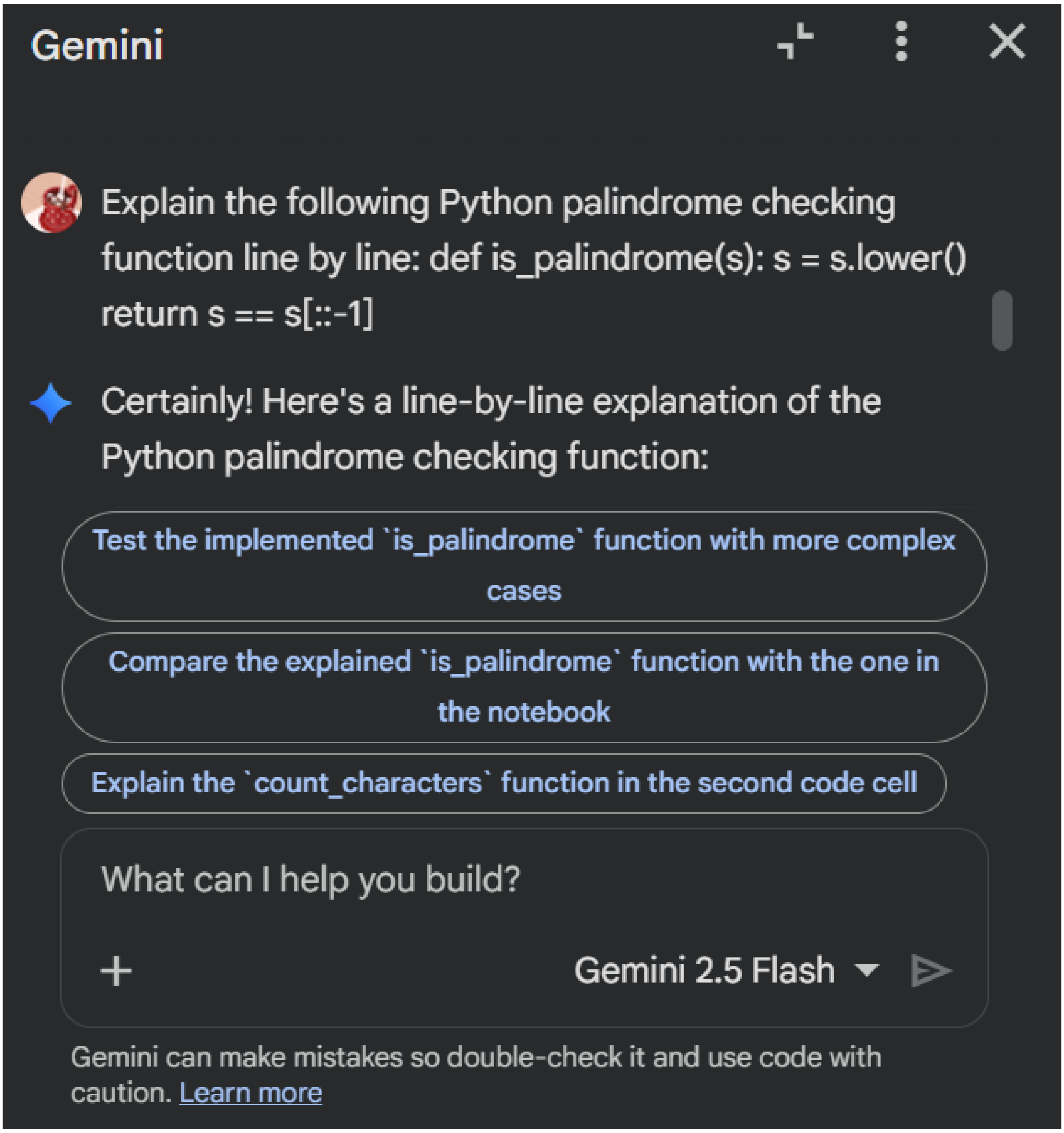
TASK 3:

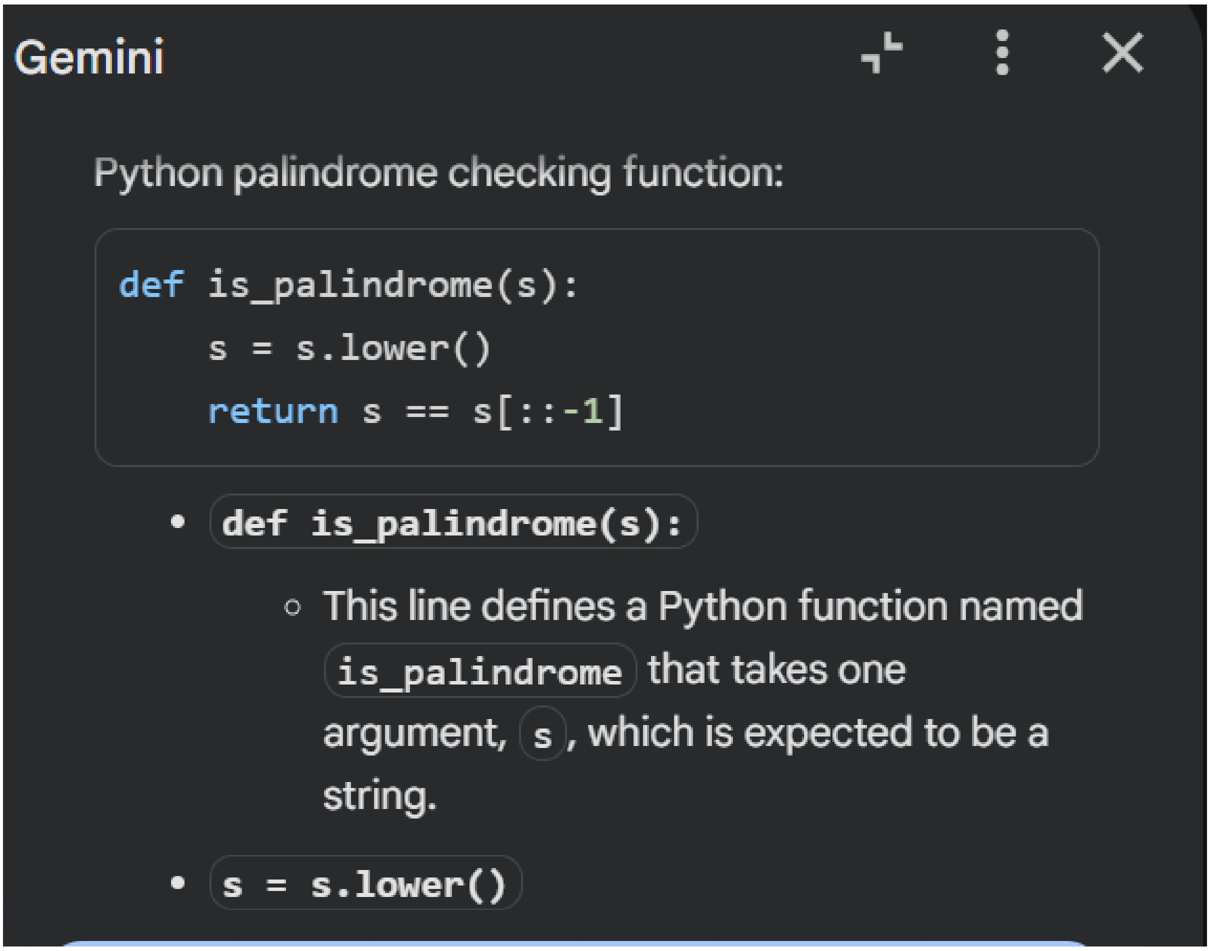
Palindrome Check – Tool Comparison

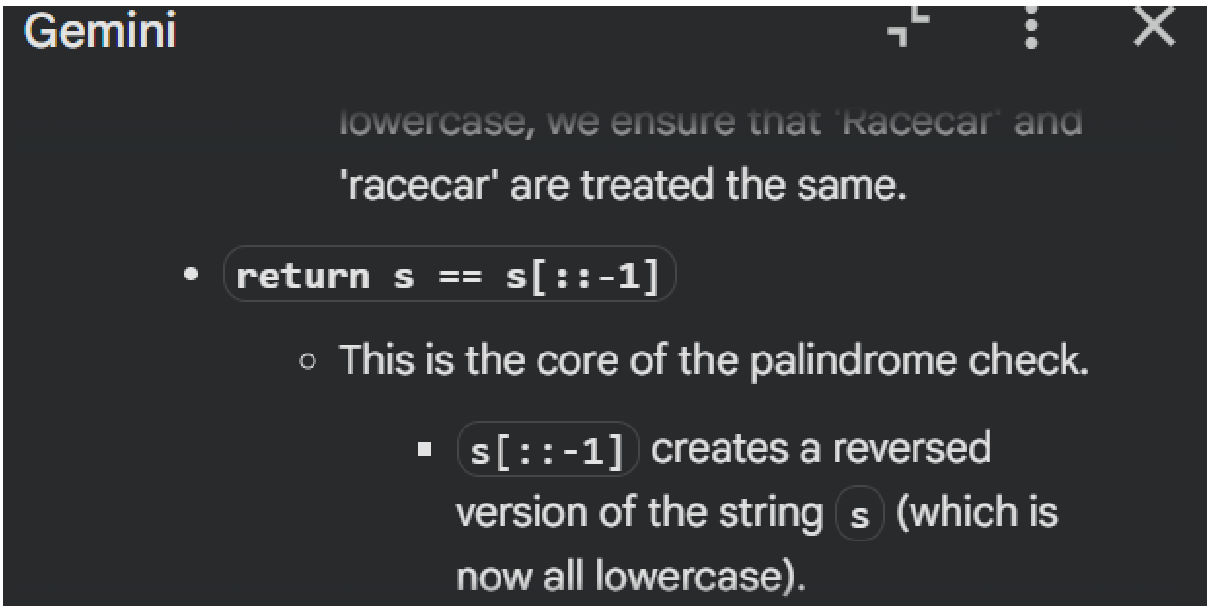
Prompt Used:

Write a Python function to check whether a string is a palindrome.

Tool	Code Quality	Readability	Comments
Gemini	Clear and beginner-friendly	High	Explains logic step-by-step
Copilot	Short and optimized	Medium	Assumes Prior Python knowledge


Gemini
Explain the following Python palindrome checking function line by line: def is_palindrome(s): s = s.lower() return s == s[::-1]
Certainly! Here's a line-by-line explanation of the Python palindrome checking function:
Test the implemented `is_palindrome` function with more complex cases
Compare the explained `is_palindrome` function with the one in the notebook
Explain the `count_characters` function in the second code cell
What can I help you build?
Gemini 2.5 Flash
Gemini can make mistakes so double-check it and use code with caution. Learn more





TASK 4

Code Explanation Using Gemini

Prompt Used:

Explain the following Python palindrome checking function line by line:

```
def is_palindrome(s):
    s = s.lower()
    return s == s[::-1]
```

The explanation helped me understand slicing operations and how reversing of a string works. Gemini explanation was clear and it was easy to follow.

CONCLUSION –

In this lab, I explored AI – ASSISTED CODING using Google Gemini in Colab. Gemini was helpful for generating beginner-friendly code with detailed explanations. While Co-Pilot provided optimised solutions. Cursor AI helped in refactoring the same. This lab helped me understand how different AI tools assist programmers in different levels.